GEBZE TECHNICAL UNIVERSITY

Computer Engineering Department

CSE 396 Module Documentation

Çılgın Kurye

Module M4: Player Client and 3D City

Company: Board&Byte **Group:** Group-8 **Date:** 7 November 2026

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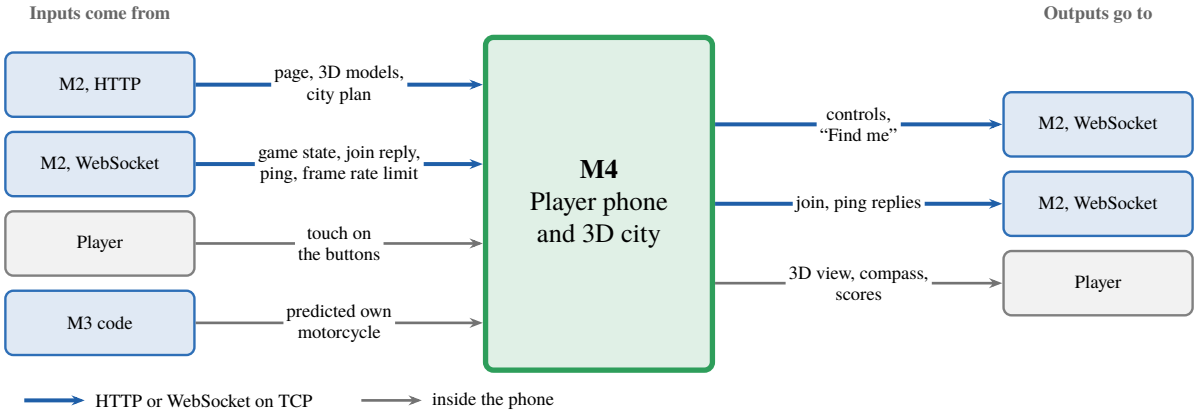
1. Purpose

M4 is the web page on each player's phone. The player scans a QR code or types the game code, chooses a team, and drives a motorcycle in a low poly 3D city with buttons on the screen. There is no map on the phone. Only a compass arrow shows the direction of the delivery point or of the store. So players must look at the LED board on the table to find their way.

Runs on	Technology	Requirements
Player phone, only the web browser, nothing to install. The page comes from M2	TypeScript web page with Three.js for 3D, low poly models from one Blender file shared with the board buildings, MessagePack messages, the M3 physics code for prediction	R1 R2 R3 R5 R7 R8 R10 R11

The phone talks only to M2. It gets the page once over HTTP and then keeps one WebSocket open. The figure shows the data in both directions.



2. Requirements covered

M4 takes part in eight requirements. The last column says what will work when M4 is finished.

ID	Requirement	What M4 does for it	When M4 is done
R1	Multiplayer game	Join, team choice, timer and the scores of all teams	A player joins and plays from the browser only
R2	Delivery cycle	Package icon and a short message when the package is delivered	The player always knows if a package is on board
R3	Driving and controls	Gas, brake, left and right buttons	The motorcycle can be driven with two thumbs
R5	No map on the phone	Only a compass arrow, and the “Find me” button	The player must use the LED board to find the way
R7	Earthquake	The camera shakes, more near the centre	An earthquake is felt on every phone
R8	Same city as the board	The 3D city is built from the shared city plan and the same models	The same building is found on the phone and on the board
R10	Network and measurement	Frame rate limit from the admin, frame rate report	The admin sees how each phone runs
R11	Engineering requirements	Own motorcycle is predicted at once and corrected by the server	Driving feels direct even on a slow network

3. Inputs and outputs

The page sends its controls on every change and at least 10 times per second, and it gets a new state 30 times per second.

Direction	Data	From / to	Content
In	Page, models, city plan	M2, HTTP	When the page loads: the program, the 3D models and the city plan
In	Game state	M2, from M1	Players, NPC cars, obstacles, scores, compass target, “Find me” waiting time, number of the last control that the server used
In	Join reply, ping, net config	M2	Player, team, colour, server time, frame rate limit
In	Touch	Player	Buttons
Out	Controls	M2, then M1	Throttle, brake, steer, a control number
Out	Join, “Find me”	M2, then M1	Player key, name, game code, team, “Find me” request
Out	Ping reply	M2	Phone time, frame rate, last state received

4. Main components

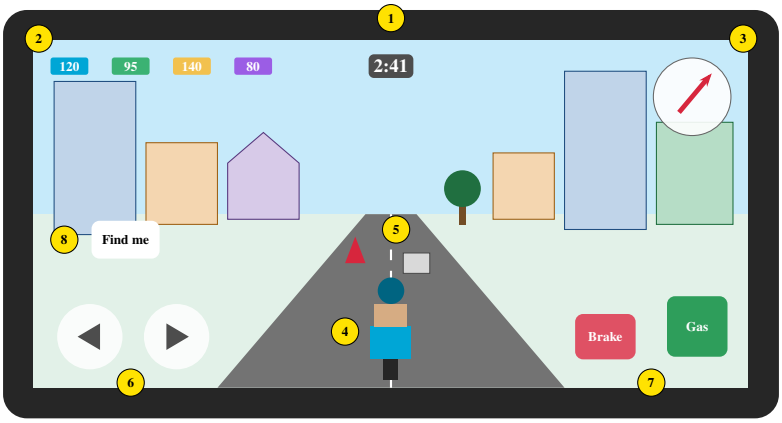
The page has five parts. The network client is the only part that talks to M2.

Component	Takes	Gives
Screens and HUD	Game state, join reply	Join, lobby, countdown, game and end screens, timer, scores, compass
Controls	Touch on the buttons	The control record and the button events
Network client	Messages from M2, the saved player key	Decoded state for the other parts, reconnect after a lost link
Prediction	Own controls, server state	Own motorcycle at once, with a smooth correction
3D city and scene	City plan, models, game state	The 3D picture with motorcycles, cars and obstacles

5. How it works

The game screen:

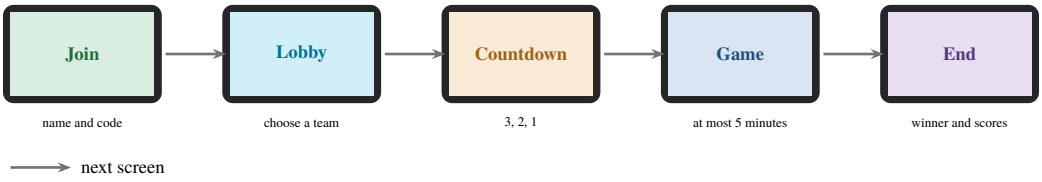
The phone is held sideways. The 3D view fills the screen, and the camera follows the motorcycle from behind. At the top the player sees the time left and the score of each team. In the top corner there is the compass. The arrow points to the delivery point, or to the store when the player has no package. It never shows the distance or the roads. The buttons are large so that the thumbs can find them without looking: left and right on one side, gas and brake on the other side. There is no delivery button: when the player stops at the delivery point, the package is delivered automatically. The "Find me" button asks the server to flash the player's LED on the board in white for 1 second. After a press, the button is grey and shows the waiting time that M1 puts into the game state.



1 Time left
2 Team scores
3 Compass arrow, no map
4 Own motorcycle, team colour
5 NPC car and an obstacle
6 Left and right
7 Gas and brake
8 "Find me" button

Screens:

The page has five screens. On the join screen the player types a name and joins the game. In the lobby the player chooses one of the four teams and waits. A 3 second countdown starts the game. The game screen is shown above. At the end, the page shows the winner and the scores of all teams.



Smooth play on a real network:

The server sends a state 30 times per second, but Wi-Fi adds delay. If the phone waited for the server, the motorcycle would react late to a button. So the phone moves its own motorcycle at once with the M3 code, and corrects it when the real state arrives. For the other players and the NPC cars, the phone keeps the last few states and draws them a little in the past. Then it can always move them smoothly between two real states, and they do not jump.

The 3D city:

The city is built from the same city plan as the LED board: 6 roads in each direction, 60 road segments and 25 blocks. Tall buildings stand in the centre, houses and trees are outside, and the four team stores in the corners have the team colours. The models are low poly and come from one Blender file. The board uses the same models for 3D printing, so a player can find the same building on the board and on the phone. Objects that repeat, like trees and houses, are drawn as instances, so a mid range phone can draw a smooth picture. During an earthquake the camera shakes, and the shaking is stronger near the centre of the earthquake.

Obstacles and falls in the 3D city:

Obstacles from the game state are drawn on their road: cones and a barrier for road work, crashed cars for an accident, slow cars for heavy traffic and a wet patch for a water puddle. Obstacles come and go during the game, and each change is seen in the 3D city in the same tick as on the board. A fallen motorcycle lies on the road, and the page shows a penalty countdown until M1 lets it stand up again.

6. Testing

M4 is tested alone against a fake server that sends recorded game states and answers the join and the controls. We check joining on Android and iPhone, every button and every screen, the compass target, the “Find me” waiting time, reconnecting after a page reload with the same player key, and the prediction with added delay and loss. We also check that a mid range phone draws the full city smoothly.